Activity 1B: Adding a Pot to Control Blinky

Purpose: To add a potentiometer to a circuit to control the speed of the blinking. Pots can also be used control volume and intensity.

Materials: Computer, Arduino Uno Board, USB cable, Breadboard, 2 coloured LED's, 200 - 330 Ω Resistor, 2 Jumper Wires, potentiometer

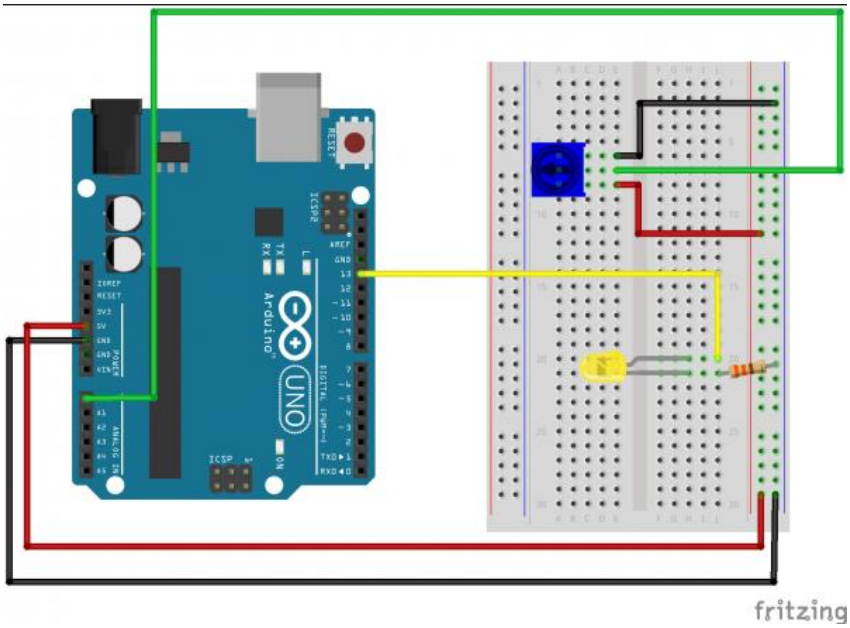
New Components:

Potentiometer (pot): A potentiometer is a 3 pin variable resistor. The middle pin outputs a voltage between 0 and 5 volts depending on the position of the knob. The more volts that the pot allows to travel through the circuit, the greater the volume, speed, or intensity of the function will be. We need to hook up the pot to one of the analog pins which are labelled A0 to A5.



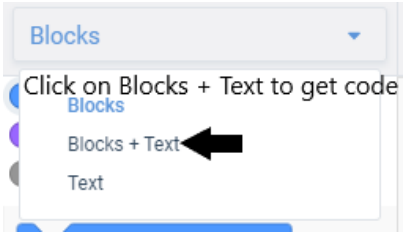
Procedure:

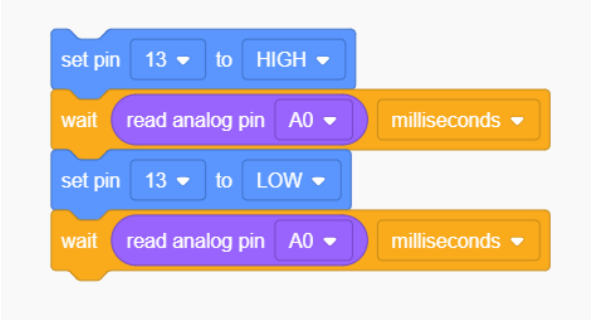
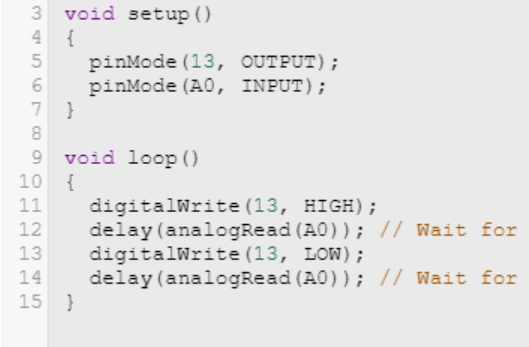
1. Wire the project using the following schematic.



2. You can create your code in the TinkerCad Editor.

3. Once in TinkerCad, simply "drag" code blocks to create your code. Put the blocks into the editor as shown below to get your light to blink in sequence: Watch the following video to assist you with this.
   VIDEO: Adding a Potentiometer to BLINKY



BLOCK CODE	ARDUINO CODE
	

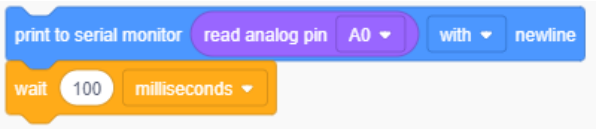
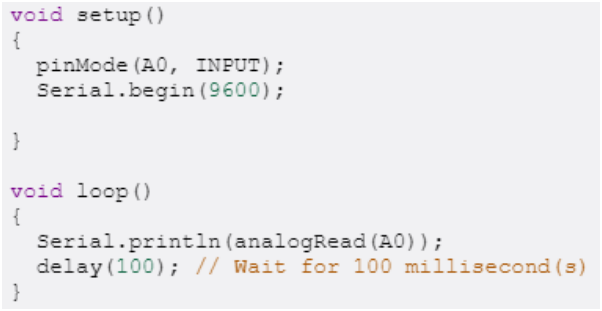

```
3  void setup()
4  {
5    pinMode(13, OUTPUT);
6    pinMode(A0, INPUT);
7  }
8
9  void loop()
10 {
11   digitalWrite(13, HIGH);
12   delay(analogRead(A0)); // Wait for
13   digitalWrite(13, LOW);
14   delay(analogRead(A0)); // Wait for
15 }
```

4. Download the Arduino Code from TinkerCad. Hook up your arduino to your computer using the USB cable and open in the Arduino Editor.

5. Your program is now loaded. Observe your LED as you turn the potentiometer and complete your worksheet titled Assignment 1B: Adding a Potentiometer to a blinking LED.

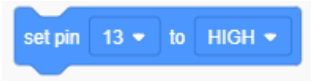
6. Now let's see if we can read the pot directly to the computer. We will use the serial monitor to "see" what the arduino senses as the potentiometer is turned. There is no need to change the wiring we will simply look at only the potentiometer and ignore the LED. Create and load the following code.

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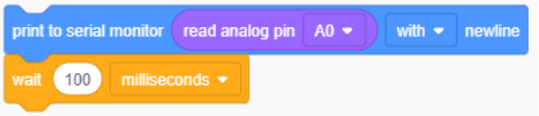
BLOCK CODE	ARDUINO CODE
	

Troubleshooting: If your circuit is not working!

1. Double check your code! Did you put in every block? Are the settings correct?
2. Refresh the website and reload the code.
3. The next step is to check your circuit!
    a. Check your wiring! Is it the same as the circuit diagram? Does the circuit go from the positive numbered pin to the ground?
    b. Next check each component. Check the led, is the long positive arm connected by wires to pin 13?
    c. Next inserting the LED directly into pin 13 and the adjacent ground. Does it light when the you code it on?



    d. Next check the individual wires in the circuit. This can be done by connecting the LED to pin 13 only then using the wire to ground the short pin. Do not connect the pin to the ground directly. (Shorting the wires) This will damage the UNO.
    e. Check the pot (potentiometer). Load this test program:



    As you turn the pot you should see the numbers change from 0 to 1023.
    f. For further information on using the serial monitor see Activity 4A Using a Serial Monitor Display.

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Assignment 1B: Adding a Potentiometer to a blinking LED.

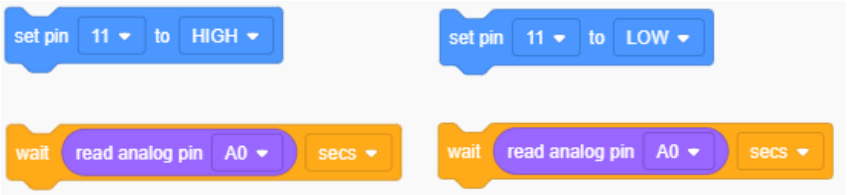
1. What does the LED do when you adjust the potentiometer? What did the serial monitor show you?

2. Look carefully at the code how did the reading from the pot change the code?

3. The pot is sometimes called a variable resistor, what is the purpose of the potentiometer? List some real life applications where a potentiometer is used or where one would be useful.

Extension Challenges:

4. Let's add a 2nd LED. Connect a 2nd LED to Pin 11.
    a. When writing code, we need to declare our pins. In the existing code, we are using pin 13 to operate the LED. Declare another LED to pin 11:
    b. Add code to make the 2nd LED light. Use the code from the existing program to help you. You will need to add the following blocks:



5. Test your code to see if both lights are controlled by the pot.

6. Change your code so that when one light flashes on the other light flashes off. You will still be able to change the speed of the flashing using your potentiometer. Test your code by running the program on your Arduino to see if when the first light flashes on, the other flashes off. (HINT: Consider the timing and how you should use your delay coding)

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a. Were you successful? Explain

b. What problems did you have to solve to make your challenge successful?

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